

Effects of Maxwell-Wagner polarization on soil complex dielectric permittivity under variable temperature and electrical conductivity

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[1] The presence of free charges and numerous discontinuities separating liquid, gas, and solid phases in partially saturated soils give rise to Maxwell-Wagner polarization that may significantly affect bulk dielectric permittivity measurements. Evidence suggests a complex interplay shaping the response of low-frequency (<100 MHz) dielectric permittivity to changes in ambient temperature and amount of ionic charges present in a wet soil. Model calculations based on the Maxwell-Wagner-Brugermann-Hanai (MWBH) theory are supported by direct measurements using a network analyzer, showing an increase in soil bulk dielectric permittivity with increasing temperature and with higher bulk electrical conductivity at low frequency range (<100 MHz). Beyond a certain frequency the decrease in permittivity of free water becomes dominant and results in a decrease in soil bulk dielectric permittivity with increasing temperature. The value of this crossover frequency can be predicted as a function of solution electrical conductivity (EC) as confirmed in limited tests. The dielectric permittivity inferred from TDR waveform travel time analysis is not significantly influenced by the low frequency range of the dielectric spectrum. In contrast, dielectric permittivity sensors operating at frequencies lower than 100 MHz are likely to show significant sensitivity to factors affecting the Maxwell-Wagner effect (temperature and electrical conductivity), hence requiring special care in measurement interpretation.

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1. Introduction

[2] Time domain reflectometry (TDR) is now considered as a standard method for repetitive and nondestructive measurement of soil water content [Topp and Ferre, 2002]. Key to the success of TDR and similar electromagnetic methods is the establishment of simple and broadly applicable relationships between measured bulk dielectric permittivity and water content, such as the empirical Topp equation [Topp *et al.*, 1980].

$$\theta = 4.3 \times 10^{-6} \epsilon_b^3 - 5.5 \times 10^{-4} \epsilon_b^2 + 2.92 \times 10^{-2} \epsilon_b - 5.3 \times 10^{-2}. \quad (1)$$

[3] Numerous applications of the Topp equation have established its usefulness for mineral soils; however, in some cases permittivity is significantly influenced by factors other than soil water content alone, resulting in significant deviations from the Topp equation [Dirksen and Dasberg, 1993; Schaap *et al.*, 1996; Tomer *et al.*, 1999].

Some of the key factors affecting permittivity measurements are electrical conductivity, temperature, soil texture, particle shape, bound water, and phase configuration [Friedman and Robinson, 2002; Jones and Or, 2002; Robinson *et al.*, 2003, 2005; Miyamoto *et al.*, 2005]. Figure 1 illustrates that the complex dielectric spectrum of liquids and other mixtures varies with frequency as a result of different relaxation processes. Most electromagnetic methods and sensors used for soil water content measurement operate at frequencies ranging from kilohertz to gigahertz in which interfacial polarization (also termed as the Maxwell-Wagner (M-W) effect) plays an important role.

[4] The M-W effect occurs in mixtures composed of segregated constituents with different dielectric permittivity and electrical conductivity. Soils represent mixtures of constituents with considerably different electrical properties (liquids, solids, air) separated by numerous interfaces. For an external electrical field applied across a soil mixture, the normal component of the electrical field must be equal across interfaces separating constituents with different dielectric properties. This requirement results in charge accumulation or interfacial charging often accompanied by considerable enhancement of bulk dielectric permittivity (higher than the sum of its components) [Alvarez, 1973; Chelidze and Gueguen, 1999; West *et al.*, 2003; Chen and Or, 2006]. Interfacial processes remain active in the frequency range of kilohertz to megahertz, well within the range of measurement of many soil water content sensors

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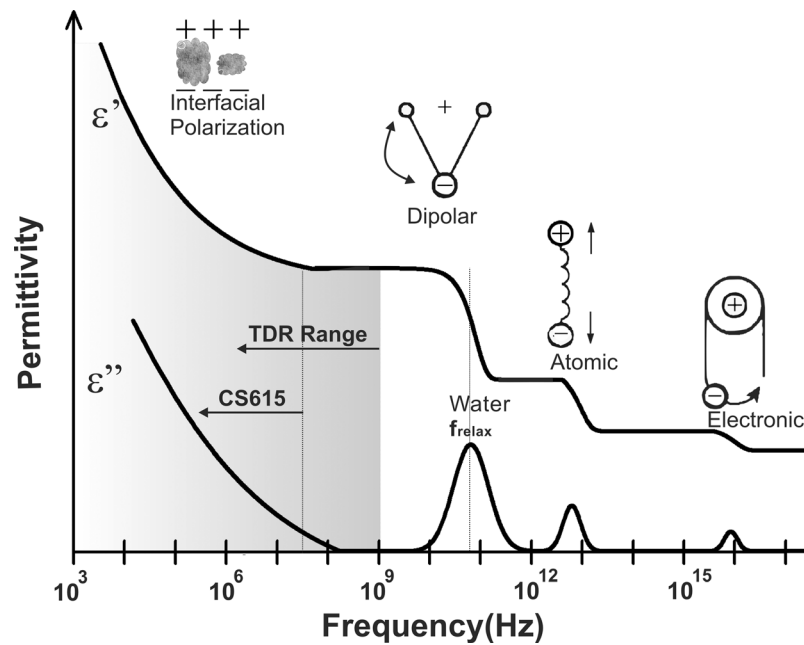


Figure 1. Schematic of the dielectric permittivity spectrum and various relaxation processes occurring at different frequencies (soil water content approximate measurement ranges of TDR and CS615 are indicated by arrows).

including TDR. The extent of M-W effect on measured dielectric permittivity is strongly dependent on soil aqueous phase electrical conductivity and ambient temperature.

[5] The primary objective of this study was to quantify the contribution of the M-W effect to soil bulk dielectric permittivity measured by TDR and other dielectric-based sensors. Our interest was motivated by the need to interpret bulk dielectric permittivity measurements and reconcile water content measurements from different sensors often operating at low frequency, where the M-W effect is dominant. Additionally, near-surface dielectric measurements of soil water content often exhibit distinct temperature effects [Halbertsma *et al.*, 1995; Wraith and Or, 1999]. These thermal effects are attributed either to reduction in dielectric permittivity of free water with increasing temperature [Pepin *et al.*, 1995] or to the release of bound water with increasing temperature [Or and Wraith, 1999], resulting in conflicting trends in bulk permittivity response. The thermal sensitivity of the M-W effect (to be presented shortly) offers an additional mechanism for thermal effects on dielectric determination of water content that must be considered. For this study we have chosen to focus on these effects on the popular TDR method; however, we offer general insights applicable to other sensors. We address these questions using a general modeling framework capable of incorporating effects of M-W polarization on dielectric permittivity under a wide range of soil textures and water contents, and considering various ambient temperatures and electrical conductivities.

[6] We begin with a brief introduction of dielectric mixing model that accounts for frequency dependency and the M-W effects on bulk dielectric permittivity. We then discuss the influence of temperature and electrical conductivity on M-W polarization, which, in turn, affects soil dielectric permittivity. We incorporate these effects into

simulated TDR waveforms used for dielectric permittivity inference and determination of water content. We compare simulation results with measurements of TDR waveforms and direct measurements of dielectric permittivity spectra. Finally, we provide a summary of key factors affecting water content measurement by TDR and other dielectric-based sensors.

2. Theoretical Considerations

2.1. Maxwell-Wagner-Bruggermann-Hanai (MWBH) and the Differential Effective Medium Approximation (DEMA)

[7] The differential effective medium approximation (DEMA) has been used to simulate bulk permittivity of porous earth materials. It incorporates effects of geometrical arrangement of various phases in the microstructures of porous media, and the effect of electrical conductivity of the aqueous phase. Details of DEMA and alternative models are described by Sen *et al.* [1981], Norris [1985], Endres and Redman [1996], Endres [2000], Asami [2002], Cosenza *et al.* [2003], and in the context of the modeling reported in this study, Chen and Or [2006].

[8] The effective permittivity according to DEMA approach is based on the following iterative steps: (1) A component of the mixture is selected as the background to which other components are added as inclusions; (2) adding infinitesimal volumes of inclusions of the other phases (with prescribed configuration) into the background matrix; (3) performing a homogenization of the new mixture; (4) using the homogenized mixture as initial background for the next step of inclusion addition; (5) repeat these steps until the target volume fraction of phases/inclusion has been reached (see details by Chen and Or [2006]).

[9] For the modeling considered in this study, we employ a simplified three-phase DEMA model for spherical inclu-

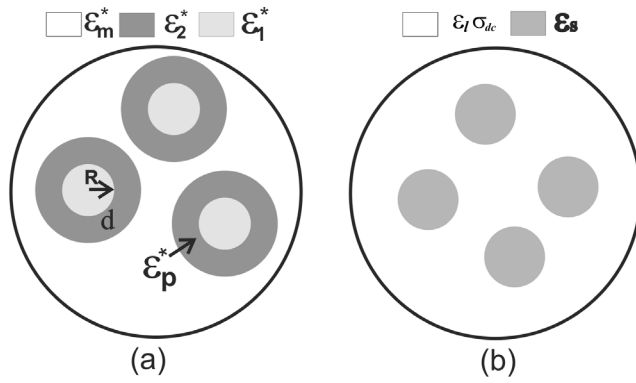


Figure 2. (a) Definition sketch of a spherical inclusion coated by a shell used for MWBH's DEMA three-phase system. (b) Wagner's two-phase model considering low inclusion concentration.

sions (equation (2)) to calculate influences of electrical conductivity, temperature, and the M-W polarization on bulk permittivity and on TDR waveforms (section 3). The high degree of saturation in most of the soil samples tested in this study simplifies our considerations to the SWA configuration only. (SWA represents the solid particles covered by water embedded in the background of air. Discussion of other phase configurations is given by *Chen and Or* [2006].)

$$\left(\frac{\epsilon_m^* - \epsilon_p^*}{\epsilon_m^* - \epsilon_2^*} \right) \left(\frac{\epsilon_m^*}{\epsilon_2^*} \right)^{1/3} = 1 - \phi$$

$$\epsilon_p^* = \epsilon_2^* \frac{2(1-\lambda)\epsilon_2^* + (1+2\lambda)\epsilon_1^*}{(2+\lambda)\epsilon_2^* + (1-\lambda)\epsilon_1^*} \quad (2)$$

$$\lambda = \left(\frac{R}{R+d} \right)^3,$$

where ϵ_m^* and ϵ_p^* are the complex dielectric permittivity of selected background matrix and the adding inclusion, which is a core material covered by a shell of another material, d is the thickness of the shell, R is the radius of the core, and ϵ_1^* and ϵ_2^* are the complex permittivities of the core and shell, respectively (Figure 2a).

2.2. Effects of Maxwell-Wagner Polarization on TDR Waveforms

[10] Detailed description of the TDR method including in-depth discussions concerning waveform analyses are

provided in a recent review by *Robinson et al.* [2003]. Here we provide a concise introduction of the method focusing on key aspects related to incorporation of the M-W effect and impact on measured waveforms essential for water content determination. Typically we measure the propagation velocity of a step voltage signal (with a bandwidth 20 kHz to 1.5 GHz for standard equipment such as the Tektronix 1502 cable tester) traveling through a waveguide (TDR probe) embedded in the porous medium. The signal travel time is a function of the permittivity of the material through which it travels. Signal travel time is inferred from a waveform recorded by the TDR marking the location (time) of the entry and reflection from the end of the probe.

[11] It is instructive to divide the elements giving rise to measured TDR waveform into three parts: the input signal, the system function (probe geometry and material), and output signal or waveform (Figure 3) [*Heimovaara*, 1994; *Friel and Or*, 1999]. It is convenient to consider the step signal generated by the instrument after traveling in the coaxial cable connected to the probe as “input signal” (assuming minimal losses in the coaxial cable [*Feng et al.*, 1999]). The system function reflects the complex interactions between the input signal probe geometry and the electrical properties of the surrounding material. The output signal or measured waveform contains information on travel time along the TDR probe as well as other influences such as attenuation due to charge migration (proportional to bulk electrical conductivity) [*Topp and Reynolds*, 1998; *Jones et al.*, 2002].

[12] For a linear time-invariant system, the output signal or the reflection waveform $r(t)$ is given by the following convolution integral:

$$r(t) = \int_{-\infty}^{\infty} v_0(t-\tau)s(\tau)d(\tau), \quad (3)$$

where $v_0(t)$ is the input signal generated by the TDR; $s(t)$ is the system function, which is dependent on the dielectric properties of the material tested (see details in Appendix A); and $r(t)$ is the reflection waveform. Conversion of these time-dependent functions into the frequency domain (using the fast Fourier transform (FFT) technique) greatly simplifies the complex convolution integral into a simple multiplication in the frequency domain as follows:

$$R(f) = V_0(f)S(f), \quad (4)$$

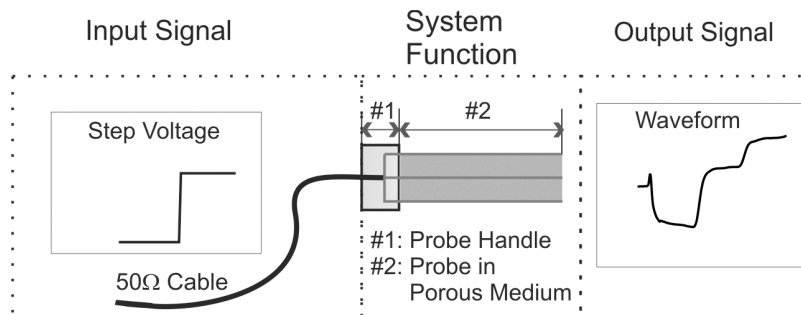


Figure 3. Illustration of TDR input signal, system function, and output signal.

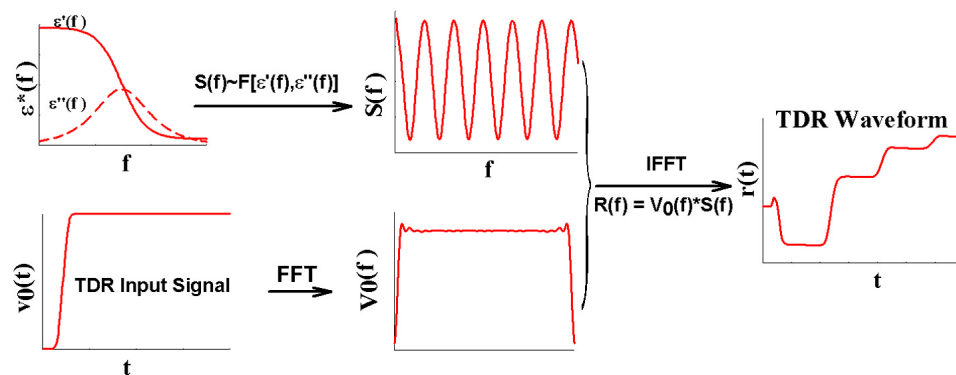


Figure 4. Schematic flowchart for TDR waveform simulation using complex permittivity, system function, and FFT/IFFT techniques.

where $R(f)$, $V_0(f)$, and $S(f)$ are the transformed waveform, input signal, and system function in the frequency domain, respectively [Friel and Or, 1999; Lin, 2003; Heimovaara et al., 2004].

[13] $S(f)$ is an explicit function of the complex dielectric permittivity of the porous medium surrounding a TDR probe (Appendix A), hence providing a means for analyzing the effects of M-W polarization on TDR waveforms. This is accomplished in the following steps:

[14] Step 1: Calculation of the bulk dielectric permittivity spectrum of the wet porous medium based on the MWBH model (equation (2)). Followed by calculation of $S(f)$ from the simulated permittivity spectrum (Appendix A, equations (A3) and (A5)).

[15] Step 2: Establishing input signal $v_0(t)$ characteristic of the TDR cable tester from either direct measurements, or from a parametric model (e.g., equation (A7)). The input signal $v_0(t)$ is then converted to its frequency domain form $V_0(f)$ using FFT.

[16] Step 3: Multiplying $V_0(f)$ and $S(f)$ as shown in equation (4) (the frequency domain equivalent to a convo-

lution integral in the time domain) to obtain the frequency domain representation of the TDR waveform $R(f)$.

[17] Step 4: Transforming the reflection signal waveform $R(f)$ back to the time domain (using inverse FFT) to obtain the simulated TDR waveform $r(t)$ which is then compared with measured-TDR waveform to assess the impact of the M-W effect on travel time analyses. A schematic flowchart is depicted in Figure 4; detailed calculations are provided in Appendix A.

3. Applications to Dielectric Permittivity Determination Using TDR

3.1. Measured and Simulated TDR Waveforms and Dielectric Spectra

[18] To test the procedure outlined in the previous section, we used a network analyzer (N-A) (Agilent 8753D equipped with high temperature dielectric probe (Agilent 85070E)) to obtain direct measurement of the dielectric spectrum of soils or known liquids. TDR waveforms were obtained from a Tektronix 1502C connected to a computer. Following tests using water and ethanol reported in Appen-

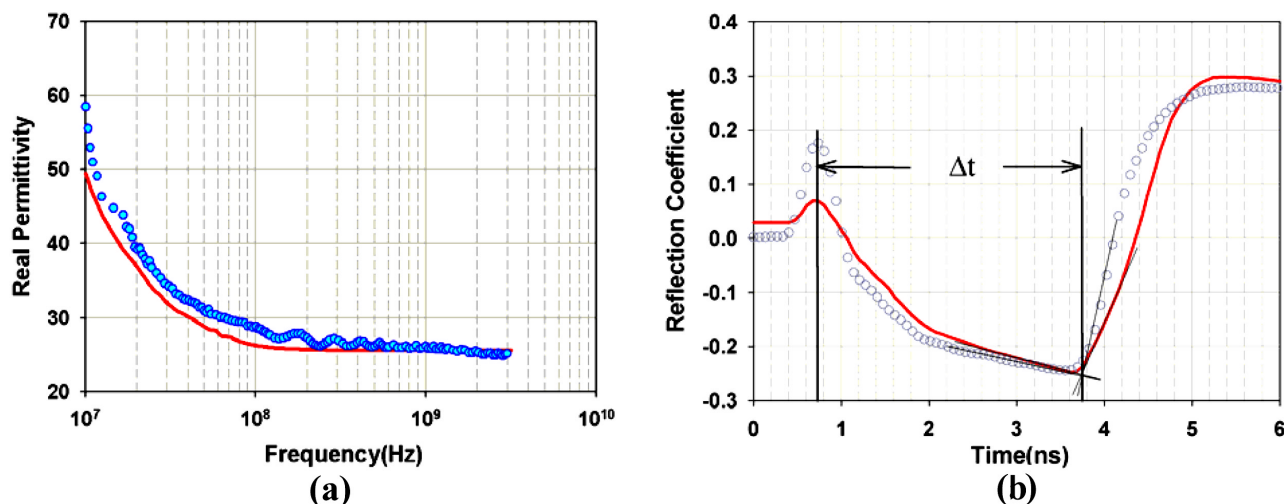


Figure 5. Comparison between measurements and simulations in Woodbridge sandy loam (around 0.44 porosity and 0.41 water content). (a) Simulated spectrum (solid line) using MWBH model versus measured data (symbols) from N-A. (b) Simulated waveform (solid line) based on spectrum simulation in Figure 5a versus TDR-measured waveform (symbols).

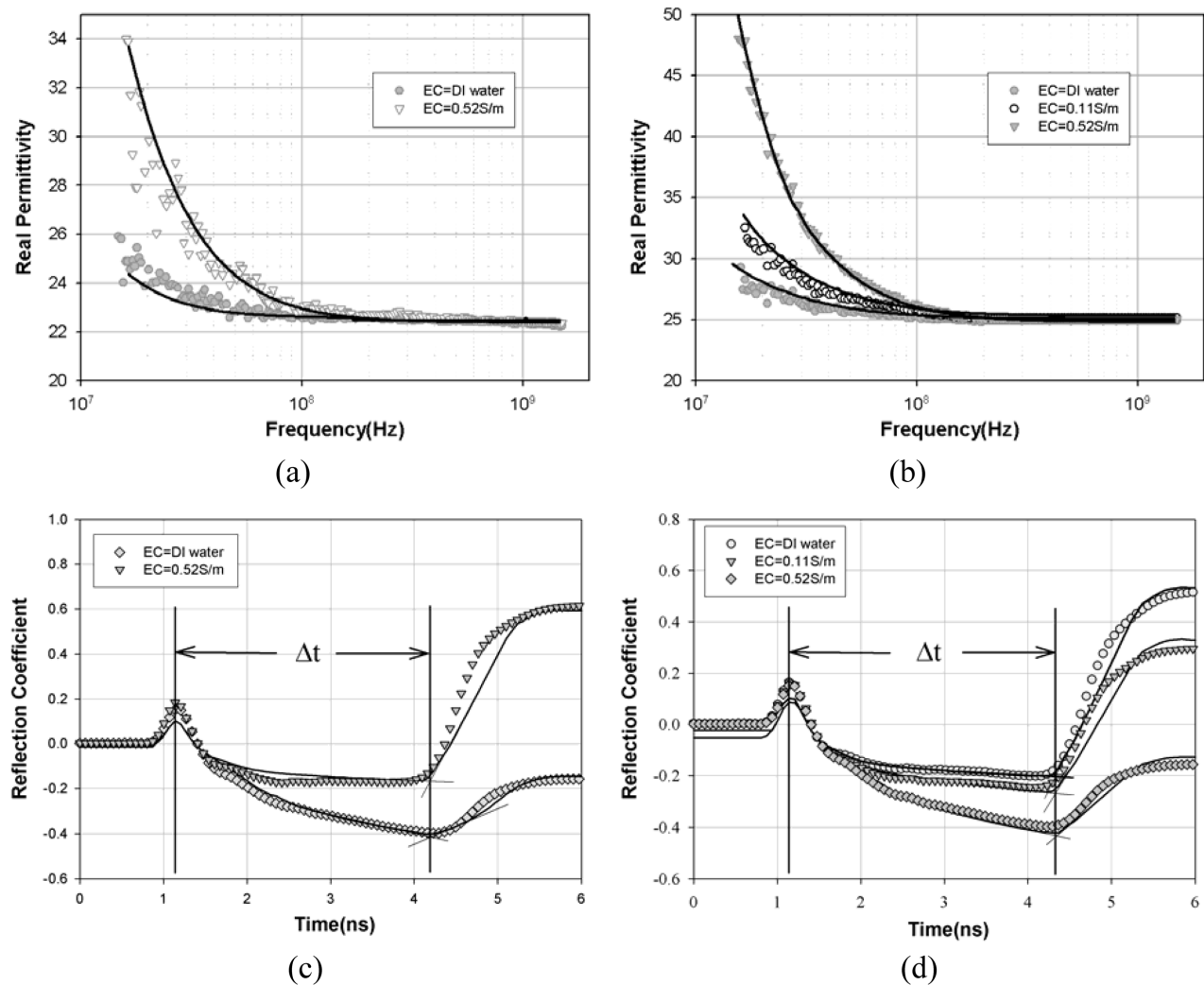


Figure 6. (a) Measured (symbols) and simulated (lines) dielectric spectra for Ottawa sand saturated with DI water and 0.54 S/m aqueous solution (porosity $\sim 38\%$; volumetric water content $\sim 34\%$). (b) Measured (symbols) and simulated (lines) dielectric spectra for Woodbridge sandy loam saturated with three aqueous solutions (DI water, 0.11 and 0.52 S/m water) of different electrical conductivities (porosity $\sim 44\%$; water content $\sim 41\%$). (c) The corresponding simulated waveforms (lines) compared with measured TDR waveforms (symbols) for Ottawa sand. (d) The corresponding simulated TDR waveforms (symbols), in media of different electrical conductivity, compared with measured TDR waveform (lines) for Woodbridge sandy loam.

dix A, we conducted tests using Woodbridge sandy loam nearly saturated with 0.125 S/m aqueous solution. The median particle size (D_{50}) was about 0.085 mm, and the porosity was 0.44 (nearly saturated water content, $\sim 0.41 \text{ m}^3/\text{m}^3$). The steps outlined in section 2.2 were followed in simulating the dielectric spectrum using the simplified version of the MWBH model equation (2) (line, Figure 5a). The modeled spectrum was used in the calculation of the system response function in the frequency domain which was then used to simulate the TDR waveform (line, Figure 5b). The measured permittivity from N-A was 22.0 (the average of the spectrum's flat region near 1 GHz). The measured permittivity from TDR waveform travel time analysis was 21.4. The resulting permittivity was relatively insensitive to frequencies in the range between 100 MHz and 1 GHz (simulated permittivity near 100 MHz ~ 21.0 , and from simulated TDR waveform ~ 20.6).

[19] The results in Figure 5 show good agreement between simulated and measured values both in the time and frequency domains, inspiring confidence in the method for simulation of waveforms in soils. These preliminary results also indicate that TDR waveform analysis is influenced by dielectric permittivity values corresponding to frequencies higher than 100 MHz (past the M-W relaxation). This important aspect will be tested in subsequent sections for different electrical conductivities and ambient temperatures using a similar modeling approach.

3.2. The Maxwell-Wagner Effect on Dielectric Permittivity and TDR Waveforms

3.2.1. Effects of the Maxwell-Wagner Polarization on Dielectric Permittivity

[20] The soil minerals are often treated as electrical insulators (nonconductive materials), and electrical conduc-

tivity is associated with properties of the soil aqueous phase. An increase in the aqueous phase electrical conductivity (EC) tends to enhance the M-W effect on bulk permittivity of the soil mixture and its frequency dependency. The EC of the soil aqueous phase was the input parameter used in our model, and it is distinguished from soil bulk electrical conductivity.

[21] Illustration of the relationships between electrical conductivity and frequency-dependent permittivity for the MWBH model (equation (2)) is complicated by the 1/3 power and integrations. Instead we demonstrate these dependencies using the simpler model of Wagner for dilute inclusion concentration (Figure 2b) considering hypothetical two-phase mixture [Sherman, 1968]. Assuming a mixture with solid inclusions that are perfect insulators (zero electrical conductivity) and ϵ_s permittivity, embedded in a background liquid with σ_{dc} electrical conductivity and ϵ_l permittivity (Figure 2b), the frequency-dependent permittivity of this mixture is explicitly given by

$$\epsilon^* = \epsilon_l \frac{2(1 - \phi_s)\epsilon_l + (1 + 2\phi_s)\epsilon_s}{(2 + \phi_s)\epsilon_l + (1 - \phi_s)\epsilon_s} + \frac{\Delta\epsilon}{1 + j\omega\tau} + \frac{\sigma^*}{j\omega\epsilon_0}, \quad (5)$$

where

$$\Delta\epsilon = \frac{9\phi_s(1 - \phi_s)\epsilon_s^2}{(2 + \phi_s)^2[2\epsilon_l + \epsilon_s + \phi_s(\epsilon_l - \epsilon_s)]} \quad (6)$$

$$\tau = \frac{1}{2\pi f_r}, \quad f_r = \frac{(2 + \phi_s)}{2\pi\epsilon_0[2\epsilon_l + \epsilon_s + \phi_s(\epsilon_l - \epsilon_s)]} \sigma_{dc} \quad (7)$$

$$\sigma^* = \frac{2(1 - \phi_s)}{(2 + \phi_s)} \sigma_{dc}, \quad (8)$$

where ϵ_0 is the dielectric permittivity of free space (8.8541×10^{-12} F/m), ϕ_s is the volumetric fraction of the solid inclusions. Here ω is the angular frequency ($\omega = 2\pi f$), $j = \sqrt{-1}$, τ is the relaxation time, and f_r is the M-W relaxation frequency. Equation (5) clearly demonstrates the dependency of bulk dielectric permittivity on relaxation frequency, which, in turn, is dependent on background solution electrical conductivity σ_{dc} (equation (7)). Similar relationships exist for the MWBH model [Hanai and Sekine, 1986], albeit the mathematical representation is more complicated.

[22] To examine the effect of different electrical conductivities, we conducted two sets of experiments measuring TDR waveforms and N-A spectra simultaneously. For the first set we used Ottawa sand (assumed spherical shape, average particle size of 0.11 mm, estimated porosity around 0.38) nearly saturated with deionized (DI) water and 0.54 S/m aqueous solution (water content for these two samples ~ 0.34 m³/m³). Modeling and measurement results are depicted in Figure 6 showing differences in the spectra for the two values of EC in the low-frequency range (less than 100 MHz) (symbols in Figure 6a for N-A measurements). In contrast, the dielectric permittivity at frequencies higher than 100 MHz was practically identical well beyond 1 GHz (in Figure 6a, N-A measured permittivity for DI

Table 1. Comparison of Dielectric Permittivity Values Deduced From Measured and Simulated TDR Waveforms (Using Travel Time Analyses) for Two Porous Media Saturated With Different Aqueous Solutions

Soil	Solution	Dielectric Permittivity	
		Measurement	Simulation
Ottawa sand	deionized water	22.68	22.35
	0.54 S/m	22.13	22.35
Woodbridge sandy loam	deionized water	23.71	24.82
	0.11 S/m	23.24	24.82
	0.52 S/m	23.47	24.82

water at 1 GHz is 22.25 and for 0.54 S/m solution is 22.30). On the basis of these input parameters, we calculated the frequency-dependent dielectric permittivity spectrum using the MWBH model (equation (2)). The simulated real part of the dielectric spectrum was in good agreement with experimental data (Figure 6a, solid lines). Simulated permittivity values were close to corresponding measured values, about 22.8 for frequencies higher than 100 MHz for both solutions.

[23] For the second set of experiments we used Woodbridge sandy loam nearly saturated with DI water, 0.11 S/m and 0.52 S/m aqueous solutions (porosity is around 0.44, water content is 0.41 m³/m³ for the three samples). The N-A measured real permittivities of the Woodbridge sandy loam saturated with DI water, 0.11 S/m, and 0.52 S/m solutions were 24.11, 23.76, and 23.78, respectively (see symbols in Figure 6b). The simulated permittivity spectra depicted in Figure 6b (lines) show reasonable agreement with measurements and converge to similar values of dielectric permittivity ~ 25.88 above 100 MHz for the three samples. Significant differences in dielectric permittivity are exhibited at the low-frequency range (<100 MHz) for the different saturating solution EC. Increasing EC results in higher dielectric permittivity at a given (low) frequency value and shifts the curve (and relaxation) to higher frequency.

[24] These effects could influence bulk dielectric permittivity measurements in porous media with similar water content but different electrical conductivities. Moreover, considering the ubiquity of free ionic charges in soils and formation of numerous additional gas-air interfaces in unsaturated soils could enhance the impact of M-W polarization on dielectric permittivity measurements with sensors operating at relatively low frequencies from kilohertz to 100 MHz. The results also suggest that in the high-frequency range (>100 MHz), the electrical conductivity's influence on the real part of the dielectric permittivity spectrum is negligible.

3.2.2. Effects of the Maxwell-Wagner Polarization on TDR Waveforms

[25] Next we examine the impact of EC and changes in the dielectric spectrum on TDR waveforms and travel time analyses. The evaluation proceeds by comparison between measured and simulated waveforms (using dielectric spectrum from MWBH model). The comparisons between measured (symbols) and simulated (lines) TDR waveforms in saturated Ottawa sand and Woodbridge SL are shown in

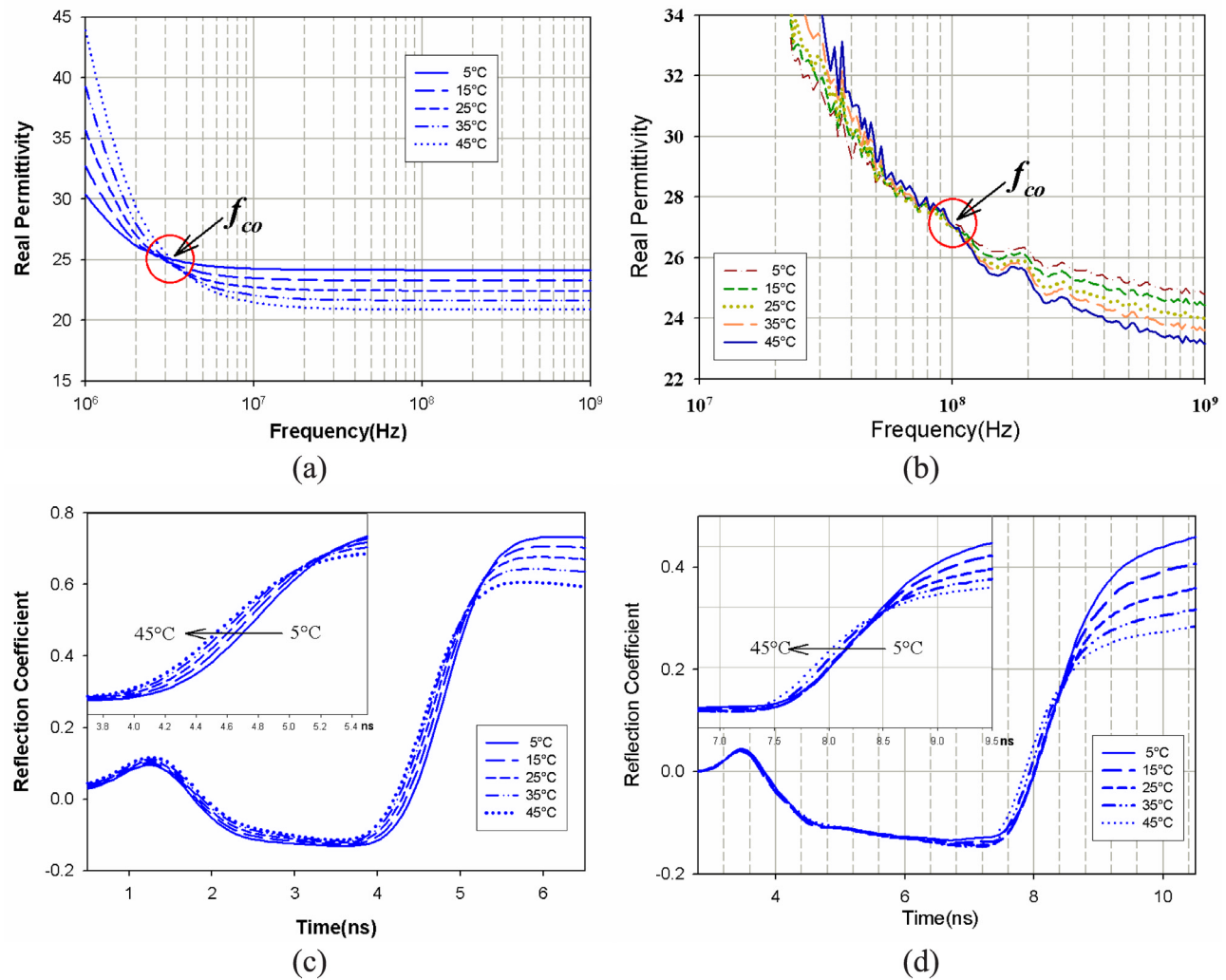


Figure 7. (a) Relative dielectric permittivity versus frequency at different temperatures (simulated by simplified spherical three-phase system; porosity is 0.4; volumetric water content is 0.35; the porous medium is assumed to be mixed with 0.05 S/m water solution). (b) Network analyzer measured real part of the dielectric permittivity spectrum of Woodbridge SL (water content $0.15 \text{ m}^3/\text{m}^3$ and solution EC 0.9 S/m) at different temperatures exhibiting crossover frequency. (c) Corresponding simulated waveform of hypothetical porous medium at different temperatures. (d) Measured TDR waveforms in Woodbridge SL at temperatures and water content corresponding to N-A measurements in Figure 7b.

Figures 6c and 6d. Resulting permittivity values from travel time analyses of these waveforms are summarized in Table 1.

[26] The results show practically no effect of EC on TDR-measured dielectric permittivity in these saturated porous samples. Despite significant differences in the dielectric spectra associated with different saturating solution EC (Figures 6a and 6b) at the low frequency, the resulting travel times were practically unaffected (yielding similar dielectric permittivity values for all EC levels; Table 1). The shapes of TDR waveforms exhibit significant effect of EC through attenuation and rounding of reflected signal; however, reflection locations determined by tangent line analyses [Wraith and Or, 1999; Robinson *et al.*, 2003] occurred at the same location on the waveform for all EC values. On the basis of results presented we conclude that features used for

travel time analyses of TDR waveforms are dominated by permittivity values at the high-frequency range ($>100 \text{ MHz}$) suggesting that M-W effect may not play an important role in TDR measurements.

3.3. Temperature Effects on Permittivity and TDR Waveforms

3.3.1. The Role of Temperature in the M-W Effect and Dielectric Permittivity

[27] Near-surface water content measurements are subject to diurnal and seasonal variations in soil temperature. Here we consider variations in the range from 5°C to 55°C . Typically, the permittivity of solid soil particles and air is not greatly affected in this temperature range, whereas soil water permittivity and electrical conductivity show much greater thermal sensitivity. The dielectric permittivity of free

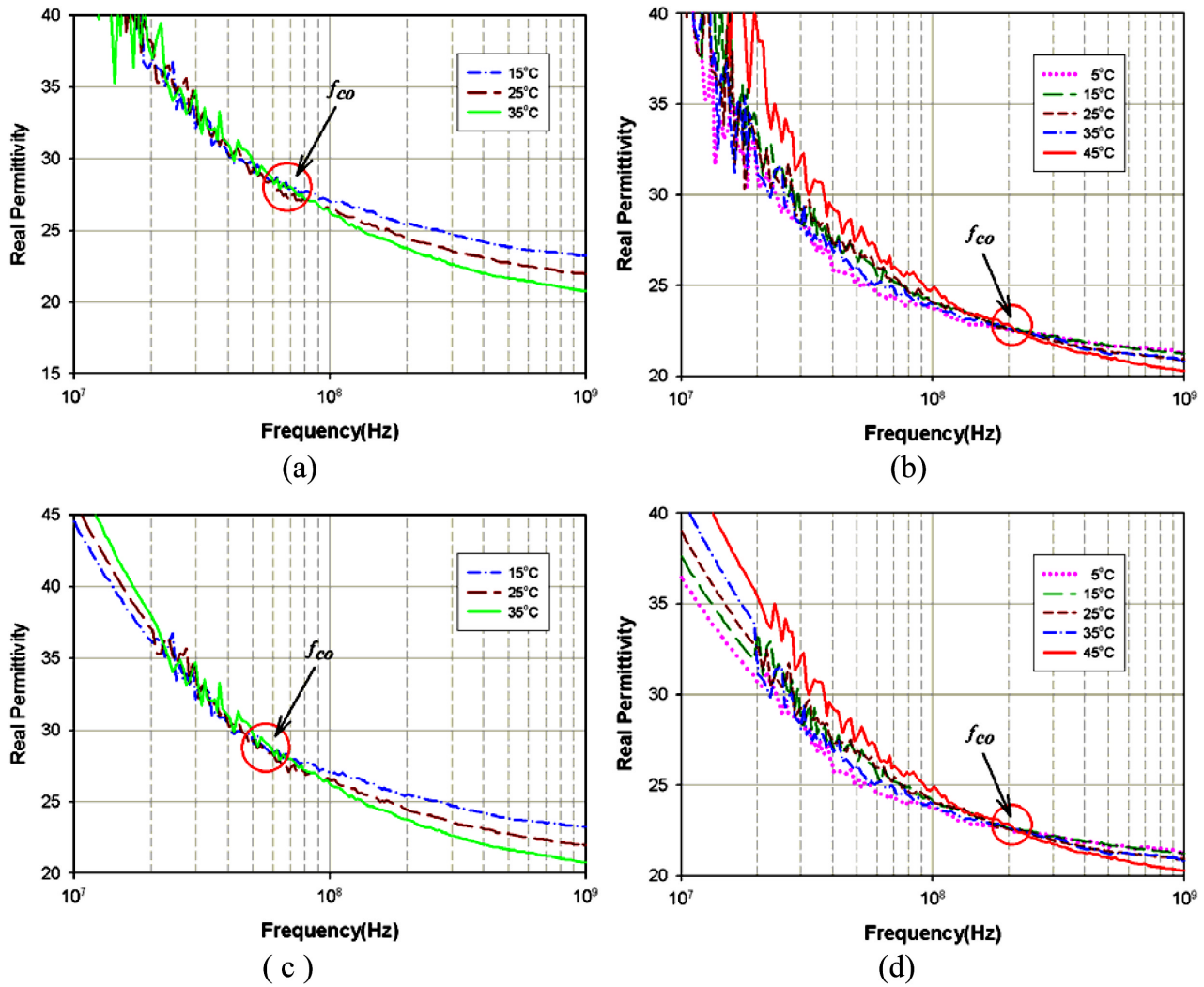


Figure 8. Network analyzer measured dielectric permittivity spectra of two soils subjected to different temperatures (measured at thermal equilibrium). (a) Millville silt loam saturated with 0.682 S/m solution. (b) Woodbridge sandy loam saturated with 1.551 S/m solution. (c) Spectrum of Millville silt loam saturated with 0.682 S/m solution after correcting electrode polarization less than 20 MHz. (d) Spectrum of Woodbridge sandy loam saturated with 1.551 S/m solution after correcting electrode polarization less than 20 MHz.

water and solution EC vary with temperature as follows [Stogryn, 1971; Weast, 1986]:

$$\epsilon'_w(T) = 87.74 - 0.4008T + 9.398 \times 10^{-4}T^2 - 1.410 \times 10^{-6}T^3 \quad (9)$$

$$\sigma_{dc}(T) = \sigma_{dc(25^\circ)} \exp[-\Delta(2.033 \times 10^{-2} + 1.266 \times 10^{-4}\Delta + 2.464 \times 10^{-6}\Delta^2)] \quad (10)$$

where T is in $^\circ\text{C}$, $\Delta = (25 - T)$, ϵ'_w is the relative permittivity of soil water, and σ_{dc} is the electrical conductivity of soil water. These relationships show a decrease in soil water permittivity with increasing temperature (hence a potential for decreasing bulk permittivity of the soil mixture [Pepin *et al.*, 1995]). However, the electrical conductivity increases with temperature (enhanced

ion mobility), thereby enhancing the M-W effect, which, in turn, could increase soil bulk dielectric permittivity at low frequency while shifting the range of M-W effect to higher frequencies (section 3.2.1). For a 1°C increase, σ_{dc} gains about 2% and ϵ'_w is reduced by 0.3%.

[28] These two competing processes were considered in the framework of the MWBH model [Hanai and Sekine, 1986]. The temperature dependencies (equations (9) and (10)) were incorporated into the calculations for a hypothetical porous medium (porosity of 0.4) “saturated” (volumetric water content of $0.35 \text{ m}^3/\text{m}^3$ to mimic experimental conditions) with aqueous solution of $\text{EC} = 0.05 \text{ S/m}$. Calculations of the dielectric spectra at the temperature range of $5^\circ\text{--}55^\circ\text{C}$ are shown in Figure 7a.

[29] The results show existence of a crossover frequency (f_{co}), below which the dielectric permittivity increases with temperature, and above which the permittivity decreases with increasing temperature. The crossover frequency

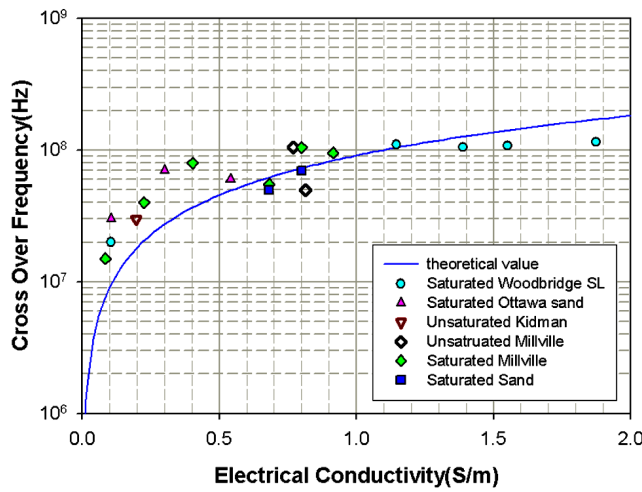


Figure 9. Crossover frequency as a function of solution electrical conductivity in a hypothetical two-phase mixture (line). Symbols represent experimental results inferred from N-A measurements at different temperatures similar to those depicted in Figure 8 (for Millville silt loam, Woodbridge sandy loam, Kidman sand, Ottawa sand, and sand).

reflects a range in the bulk permittivity spectrum where the increasing permittivity due to M-W effect of the mixture is balanced by the decrease in dielectric permittivity of free water.

[30] To further investigate this thermal effect, we conducted an experiment using Woodbridge sandy loam wetted with 0.9 S/m solution to volumetric water content of $0.15 \text{ m}^3/\text{m}^3$. The wet soil sample was allowed to equilibrate and brought to different temperatures using a temperature bath (allowing 24 hours for thermal equilibration). N-A measurements of the real part of dielectric permittivity spectrum of the soil sample confirmed model predictions as seen in Figure 7b with crossover frequency near 90 MHz.

[31] Further, we conducted a series of experiments using saturated Millville silt loam ($\text{EC} = 0.0846, 0.225, 0.403, 0.682, 0.8, \text{ and } 0.915 \text{ S/m}$) (Figure 8a), Woodbridge sandy loam saturated with different water solutions ($\text{EC} = 0.105, 1.146, 1.389, 1.551, \text{ and } 1.873 \text{ S/m}$) (Figure 8b), and also Ottawa sand, Kidman sand, and unsaturated Millville silt loam with different solutions. For these measurements, we also corrected for electrode polarization (Figures 8c and 8d). Appendix B shows that electrode polarization becomes negligible at frequencies higher than $\sim 20 \text{ MHz}$. The measurement results indicate different crossover frequencies corresponding to different solutions ECs.

[32] The opposing trends in thermal effects on free water dielectric permittivity versus impact on M-W polarization may result in a complex response in soil bulk dielectric permittivity as a function of temperature and frequency. To describe the onset and location of the crossover frequency as a function of key parameters such as solution EC and temperature, we seek an analytical expression using the simplified Wagner's model (equations (5)–(8)).

[33] The crossover frequency (f_{CO}) defines a frequency value (or a narrow range) below which the bulk dielectric permittivity increases with temperature increase, and above which the dielectric permittivity decreases as temperature increases. Mathematically, the crossover frequency is de-

fined as the frequency where the derivative of permittivity with respect to temperature equals zero ($\partial\epsilon/\partial T = 0$). Introduction of the thermal dependencies (equations (9) and (10)) into equation (5), and taking the partial derivatives with respect to temperature, and equating to zero yields the following closed-form expression for the crossover frequency as a function of key input parameters for two-phase mixtures (e.g., soil saturated with saline solution):

$$f_{CO} = \sqrt{\frac{-B + \sqrt{B^2 - 4AC}}{2A} - f_{M-W}^2} \quad (11)$$

where

$$\begin{aligned} -B &= \frac{9\phi_l(1-\phi_l)\epsilon_s^2\sigma_{dc}[-(1-\phi_l)(\phi_l+3)\sigma_{dc}\frac{\partial\epsilon_l}{\partial T} - 2D\frac{\partial\sigma_{dc}}{\partial T}]}{(2\pi\epsilon_0)^2D^4} \\ A &= \frac{9\phi_l\epsilon_s^2}{D}, \quad D = (2+\phi_l)\epsilon_s + (1-\phi_l)\epsilon_l \\ C &= -2\frac{9\phi_l(1-\phi_l)\epsilon_s^2\sigma_{dc}^3\left[(1-\phi_l)D\frac{\partial\sigma_{dc}}{\partial T} - (1-\phi_l)^2\sigma_{dc}\frac{\partial\epsilon_l}{\partial T}\right]}{(2\pi\epsilon_0)^4D^6} \\ \frac{\partial\epsilon_l}{\partial T} &= -0.4008 + 1.88 \times 10^{-3}T - 4.23 \times 10^{-6}T^2 \\ \frac{\partial\sigma_{dc}}{\partial T} &= \sigma_{dc(25^\circ)} \exp[-\Delta(2.033 \times 10^{-2} \\ &\quad + 1.266 \times 10^{-4}\Delta + 2.464 \times 10^{-6}\Delta^2)] \\ &\quad * (2.033 \times 10^{-2} - 2.532 \times 10^{-4}\Delta - 7.392 \times 10^{-6}\Delta^2) \\ f_{M-W} &= \frac{(1-\phi_l)\sigma_{dc}}{2\pi\epsilon_0[(2+\phi_l)\epsilon_s + (1-\phi_l)\epsilon_l]} \end{aligned} \quad (12)$$

The resulting expression for f_{CO} (equation (11)) is a strong function of solution EC as shown in Figure 9 (line) for a hypothetical two phase mixture (i.e., saturated soil) with solids volume fraction of $\phi_s = 55\%$ (or $\phi_l = 0.45$), and solid permittivity $\epsilon_s = 4.6$. We conducted a series of tests of this approximation using different soil samples (Millville silt loam, Woodbridge sandy loam, Kidman sand, Ottawa sand, and sand) saturated with a broad range of solution EC (NaCl aqueous solutions). The samples were brought to thermal equilibrium using a temperature-controlled water bath and were measured with a network analyzer to obtain their dielectric permittivity spectra. Compilation of these experimental results and visual interpretation of f_{CO} from spectra of a sample under different temperature (similar to results in Figure 8) provide a general agreement with trends predicted by the model (equation (11)) even for a few unsaturated (three-phase) samples as seen results (symbols) depicted in Figure 9. The resulting experimental crossover frequency (f_{CO}) tends to level off at frequencies around 100 MHz for a wide range of solution EC values.

[34] The analyses and measurements demonstrate a well-defined low-frequency range in which the M-W effect dominates the behavior of frequency-dependent bulk dielectric permittivity, and a crossover range where high-frequency behavior sets in. The high-frequency range is characterized by decreasing permittivity with increasing temperature and convergence of the dielectric permittivity to a nearly constant value (virtually no frequency dependency). For simplicity, we do not consider other thermal effects that might take place at the high-frequency range

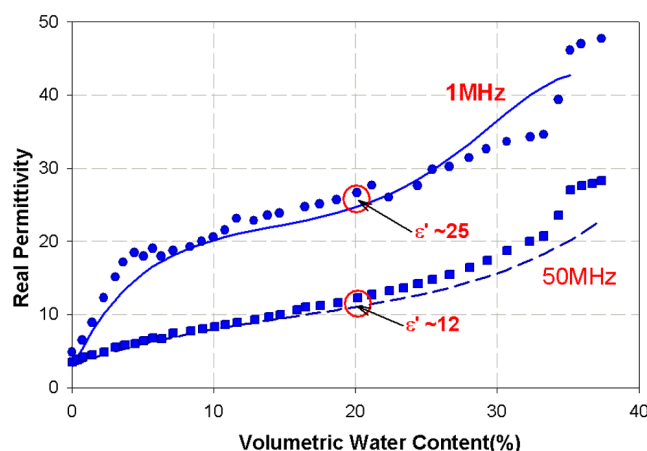


Figure 10. Measured (symbols) and simulated (lines) soil bulk permittivity values of Wilder silt soil as function of volumetric water content at two measurement frequencies of 1 and 50 MHz [Campbell, 1990].

such as release of bound water with increasing temperature [Or and Wraith, 1999], which should affect both N-A and TDR measurements. The value of critical frequency which separates the low- and high-frequency behaviors varies primarily with solution electrical conductivity (there is a weak dependency on water content not explicitly accounted for in the two-phase model). On the basis of these results we would expect that the thermal response of water content sensors operating at the low-frequency range (<100 MHz) would probably exhibit an increase in dielectric permittivity with increasing temperature due to the M-W effect. Conversely, sensors operating at the high frequency (>100 MHz) would show a decrease in dielectric permittivity with increasing temperature, especially in soils with coarse texture and low surface area where bound water effects are minimal.

3.3.2. Effects of Temperature and M-W Polarization on TDR Measurements

[35] To analyze the combined thermal and M-W polarization effects on TDR waveforms, we employ the procedure outlined in section 2.2 to simulate TDR waveforms for a hypothetical porous medium (section 3.3.1) considering different temperatures. The simulated waveforms (Figure 7c) are based on the corresponding dielectric permittivity spectra shown in Figure 7a. Although dielectric permittivity spectra exhibited different trends with increasing temperature, the effect of temperature manifested in simulated TDR waveforms shows only a decrease in travel time with increasing temperature (noting that signal attenuation was affected). This suggests that TDR waveform features used for travel time analyses (i.e., locations of reflections) are determined primarily by high-frequency components of the dielectric spectrum (>100 MHz). Similar results were obtained for Woodbridge sandy loam comparing N-A measured spectra at different temperatures (Figure 7b) and the corresponding measured TDR waveforms (Figure 7d) showing shorter travel time with increasing temperature. In other words, both simulated and measured TDR waveforms show that while attenuation increases with increasing temperature (effective EC increases), the signal travel time decreases with increasing temperature,

confirming the dominance of the high-frequency behavior on TDR waveforms.

[36] Summarizing, the thermal response of complex bulk permittivity is influenced by the interplay between effects of EC and temperature on enhancing the M-W effect at the low-frequency range, and reducing dielectric permittivity at higher frequency (due to the effect on free water permittivity). These factors affect travel time analyses deduced from TDR waveforms at frequencies higher than the crossover frequency (>100 MHz) despite the inherently broad frequency content of TDR signals (20 KHz to 1 GHz).

3.4. Effects of Maxwell-Wagner Polarization on Other Dielectric-Based Sensors

[37] The foregoing discussion suggests that sensor measurement frequency is a critical factor in interpretation of dielectric measurements and conversion to water contents, particularly at low frequencies. To illustrate this aspect, we use Campbell's [1990] dielectric measurements in soils of different textures obtained at frequencies of 1, 5, and 50 MHz using a network analyzer and coaxial sample holder. For the fine-textured soils (silt and clay soils), measured dielectric permittivity exhibited substantial differences between the various frequencies. For example, Campbell's [1990] measurements (symbols) for Wilder silt soil obtained at two frequencies are depicted in Figure 10. This example illustrates that a volumetric water content of $0.20 \text{ m}^3/\text{m}^3$ would correspond to a dielectric permittivity value of ~ 25 for 1 MHz and to only ~ 12 at 50 MHz! Alternatively, a measured dielectric permittivity value of 20 with a sensor operating at 1 MHz would correspond to volumetric water content of $0.10 \text{ m}^3/\text{m}^3$, whereas for a sensor operating at 50 MHz this value would correspond to $0.33 \text{ m}^3/\text{m}^3$.

[38] These differences highlight the need for a model that is capable of accounting for low-frequency behavior (the M-W effect) and for providing a physically based link between dielectric permittivity and water content for a given sensor frequency. We used the MWBH model to simulate the dielectric permittivity at different frequencies (Figure 10, lines, using a combination of AWS and SWA phase configurations as described by Chen and Or [2006]). Despite limitations of the modeling results, the simulated lines capture the main trends exhibited by measurements and account for the M-W effect. Results from numerous N-A measurements conducted in our lab using different soil types and EC conditions (not reported) show typical dispersive spectra associated with the M-W effect at frequencies up to ~ 100 MHz (e.g., Figures 6a and 6b) within which many new dielectric sensors operate. In addition to large changes in frequency-dependent relationships between dielectric permittivity and water content at low frequencies (Figure 10), field applications are further complicated by thermal sensitivity discussed above, often requiring site specific calibration [Chandler et al., 2004], and are subject to ongoing debate [Nadler, 2005].

4. Summary and Conclusions

[39] The main focus of this study was on the role of the M-W effect associated with inhomogeneous mixtures of constituents with different dielectric properties on bulk dielectric permittivity measurements. A framework based on the MWBH formalism was used to model frequency-

dependent dielectric permittivity of soils subjected to different ambient temperatures and containing aqueous solutions with different values of electrical conductivity. Through the effects on frequency-dependent dielectric permittivity, we were able to examine the role of the M-W effect on TDR waveforms (see Appendix A). The results showed that despite significant effect of M-W polarization on dielectric permittivity at lower frequencies, the effect on travel time analyses of TDR waveforms was negligible.

[40] The primary variable affecting the magnitude and frequency of occurrence of the M-W effect within a dielectric permittivity spectrum is the solution electrical conductivity as illustrated analytically and experimentally using network analyzer measurements (Figures 6a and 6b). However, the effect of EC on TDR waveforms was limited to signal attenuation, with little impact on travel time analyses used to deduce bulk dielectric permittivity and water content (Figures 6c and 6d).

[41] Analyses of thermal effects on M-W polarization yielded an interesting interplay between amplification of the M-W effect with increasing temperature at low frequency, and monotonic decrease in dielectric permittivity of free water dominating the mixture at higher frequencies. Theoretical predictions were supported by network analyzer measurements of soil dielectric spectra for different temperatures, confirming the existence of a crossover frequency that separates these two trends. We used Wagner's two-phase dielectric mixing model to develop a closed-form expression for the crossover frequency (equation (11)). Tests using soil samples saturated with solutions of different EC resulted in reasonable agreement with model prediction of crossover frequency (Figure 9). The existence of a predictable crossover frequency offers a convenient and physically based criterion for separating between the low- and high-frequency measurement ranges with respect to dielectric measurements of soil water content.

[42] The negligible impact of the M-W effect on TDR travel time analyses suggests that the effective TDR frequency range is higher than 100 MHz, and highlights the need for frequency-dependent calibration for interpretation of dielectric measurements using water content dielectric sensors operating at the low-frequency range.

Appendix A: Simulation of TDR Waveform

A1. System Response or the S_{11} Function

[43] The TDR waveform is represented as a convolution integral:

$$r(t) = \int_{-\infty}^{\infty} v_0(t - \tau) s(\tau) d(\tau), \quad (A1)$$

where $v_0(t)$ is the input signal, $s(t)$ is the system function, affected by probe geometry and the material contained in the probe, and $r(t)$ is the reflection waveform.

[44] Conversion of equation (A1) to the frequency domain simplifies the convolution integral into a simple multiplication:

$$R(f) = V_0(f) S(f) \quad (A2)$$

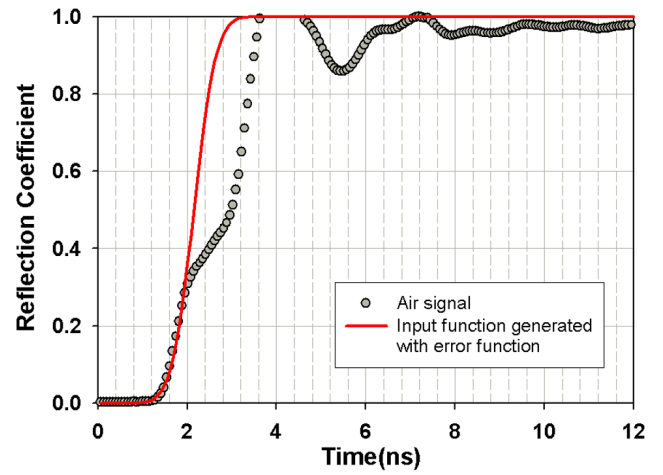


Figure A1. The input signal generated by error function (line) is exactly the same as the tangential curve of the air waveform (points).

[45] The reflection coefficient from the probe (including effect of probe handle) is given by [Heimovaara, 1994; Feng et al., 1999; Friel and Or, 1999]

$$S_{11} = \frac{\rho_e(f) + H(f, 2L)}{1 + \rho_e(f)H(f, 2L)}, \quad (A3)$$

where

$$\begin{aligned} \rho_e(f) &= \frac{z_p - z_h}{z_p + z_h} \\ H(f, 2L) &= \exp[-\alpha(f) * 2L + j\beta(f) * 2L] \\ \alpha(f) &= \frac{2\pi f}{c} \sqrt{\frac{\epsilon'_{s_mix}(f)}{2} \left[\sqrt{1 + \left(\frac{\epsilon''_{s_mix}(f)}{\epsilon'_{s_mix}(f)} \right)^2} - 1 \right]} \\ \beta(f) &= \frac{2\pi f}{c} \sqrt{\frac{\epsilon'_{s_mix}(f)}{2} \left[\sqrt{1 + \left(\frac{\epsilon''_{s_mix}(f)}{\epsilon'_{s_mix}(f)} \right)^2} + 1 \right]} \\ z_p &= \frac{60}{\sqrt{\epsilon'_{s_mix}(f)}} \ln\left(\frac{b}{a}\right), \quad z_h = \frac{60}{\sqrt{\epsilon'_h(f)}} \ln\left(\frac{b}{a}\right), \end{aligned} \quad (A4)$$

z_p and z_h are the impedances of probe and probe handle; b is the inner diameter of the outer conductor of probe, a is the outer diameter of the inner conductor of probe; L is the length of the probe; c is the speed of light in vacuum; and $\epsilon'_{s_mix}(f)$ and $\epsilon''_{s_mix}(f)$ are the real and imaginary components of soil permittivity.

[46] The reflection coefficient at the interface between the coaxial cable and the probe's handle is

$$S(f) = \frac{\rho_h(f) + S_{11} * H(f, 2L')}{1 + \rho_h(f) * S_{11} * H(f, 2L')}, \quad (A5)$$

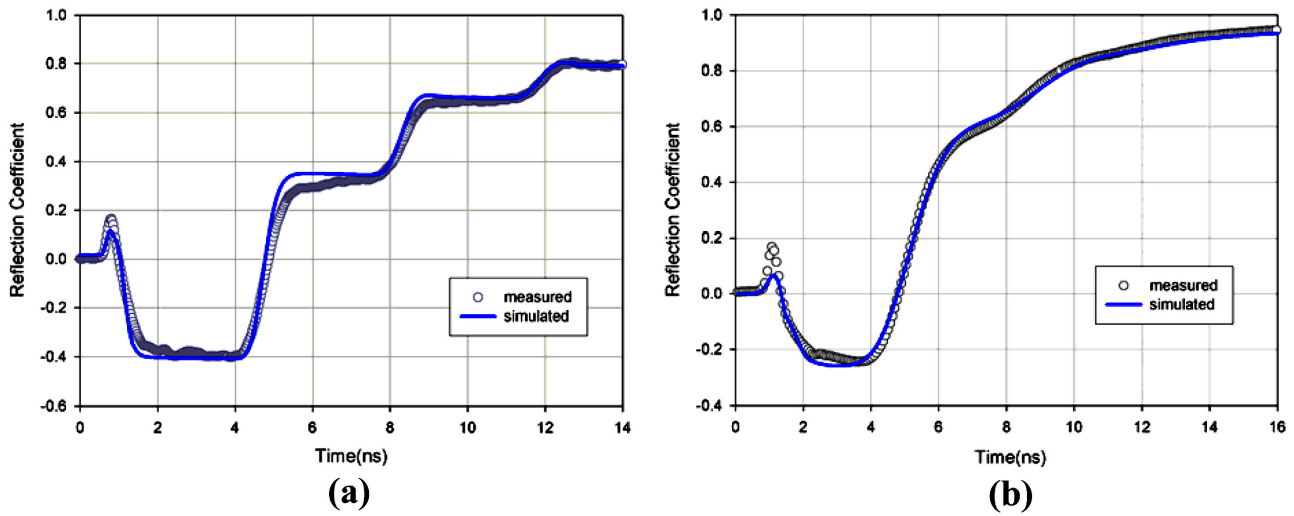


Figure A2. FFT simulated waveforms versus TDR-measured waveforms. (a) Simulated water waveform (solid line) versus measured data (symbols) from TDR (probe length = 0.058 m). (b) Simulated ethanol waveform (solid line) versus TDR-measured waveform (symbols) (probe length = 0.101 m).

where

$$\rho_h(f) = \frac{z_h - z_0}{z_h + z_0}$$

$$H(f, 2L') \approx \exp \left[j2\pi f \sqrt{\epsilon'_h(f)} * 2L'/c \right]. \quad (\text{A6})$$

Z_0 is the characteristic impedance of the coaxial cable connected to the cable tester (typically $\sim 50 \Omega$); $\epsilon'_h(f)$ is the real part of permittivity of probe handle (imaginary part of the probe handle is negligible); L' is the length of probe handle.

A2. Input Signal

[47] We may obtain the input signal by removing the center conductor of a TDR probe; by taking a tangent to the fast rise time portion of a waveform measured in air [Jones and Or, 2004]; or by parameterization of the input signal using an arbitrary function such as the error function [Heimovaara et al., 2004]:

$$v_{0,m}(t) = \frac{1 + \text{erf}[\alpha(t - t_0)]}{2}, \quad (\text{A7})$$

where erf is the error function, t represents the time, α is a fitting parameter, and t_0 is the position where the input signal starts. In this paper we used equation (A7) to produce the input signal that was practically identical to a tangent method proposed by Jones and Or [2004] (Figure A1).

A3. Simulated Water and Ethanol Waveforms

[48] Following the steps outlined in the flowchart in Figure 4, we first applied the method to simulate waveforms of known liquids such as deionized (DI) water and ethanol (room temperature around 24°C). Simulated waveforms were in excellent agreement with TDR measured waveforms in the same liquids as illustrated in Figures A2a and A2b.

[49] The scheme and these preliminary tests paved the way for incorporating frequency-dependent dielectric permittivity spectra through the $S(f)$ function for soils with different solution EC and subjected to different ambient temperatures and the evaluation of impacts on waveforms and travel time analyses.

Appendix B: Correction for Electrode Polarization

[50] Figure B1 shows the equivalent circuit for the network analyzer cable-probe-material system. Z_p is impedance due to electrode polarization in series with Z_s , the impedance of measured material. R , L , and C_f are electric parameters (resistance, inductance, and capacitance) associated with the network analyzer machine and the open-ended dielectric probe. For the open-ended coaxial probe geometry used in this study, the capacitance C_f is in parallel with Z_p and Z_s . However, the parasitic parameters R and L are related to the signal traveling from the source to the probe, and hence these are considered in series with the probe and the dielectric material at the probe's surface [Stuchly and Stuchly, 1980; Bordi et al., 2001].

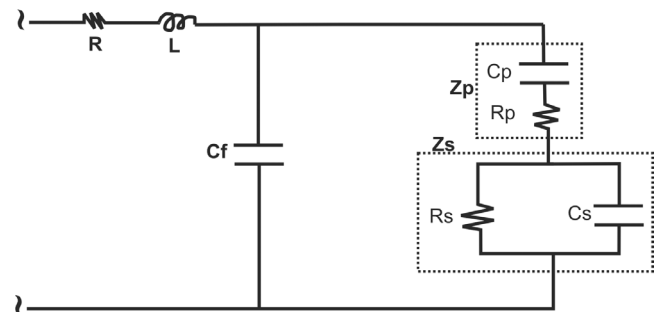


Figure B1. Equivalent circuit NA probe and associated electrode polarization elements [Bordi et al., 2001].

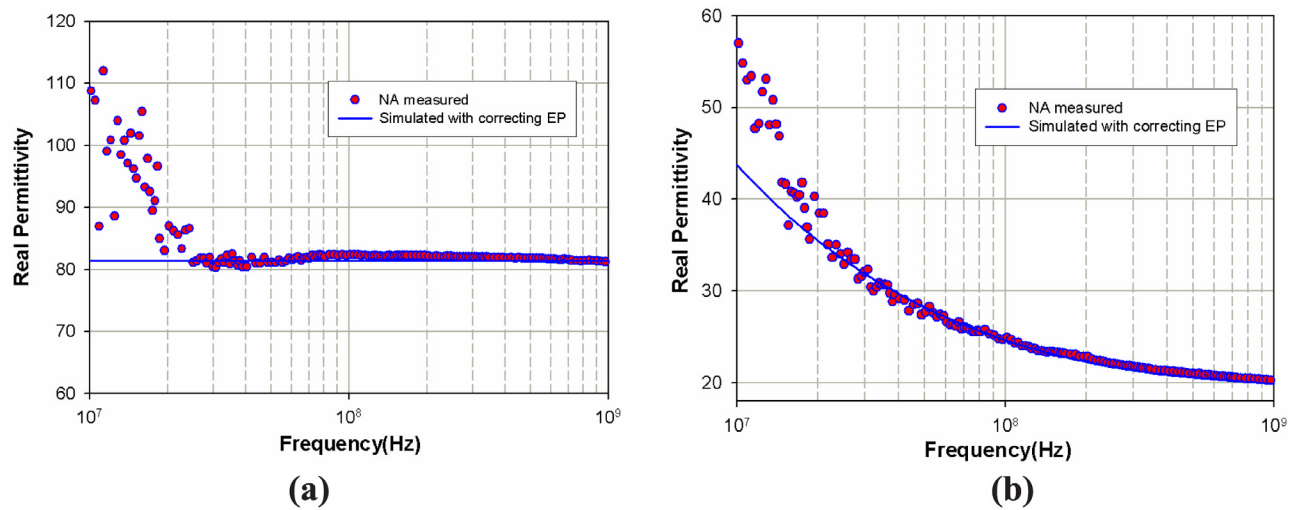


Figure B2. Corrected spectra versus NA measured spectra. (a) Corrected water solution spectrum (solid line) versus measured data (symbols) from NA at 15°C (EC = 0.67 S/m). (b) Corrected Woodbridge sandy loam spectrum (solid line) versus NA-measured spectrum (symbols) at 45°C (saturated, mixed with 1.551 S/m water solution).

[51] The correction for electrode polarization follows a procedure proposed by Bordini *et al.* [2001]:

[52] Step 1: Calibrate the probe with standard and known dielectric to establish system parameters R , L , and C_f . The known material should be homogeneous and induce no electrode polarization (DI water, dry solid, etc.).

[53] Step 2: Calculate the impedance due to electrode polarization by the constant phase angle (CPA) method. This CPA method assumes that the interface double layer impedance can be replaced by a constant-phase-angle element whose impedance has a power law singularity to ω [Liu, 1985; Macdonald, 1987; Roldán-Toro and Solier, 2004]:

$$Z_p = K^{-1} (j\omega)^{-\alpha}, \quad (\text{B1})$$

where K is a constant and $0 \leq \alpha \leq 1$, and K is

$$K = f_g \sigma^{1-\alpha}, \quad (\text{B2})$$

where f_g is a constant related with the geometry of probe and σ is the electrical conductivity of solution.

[54] Step 3: Subtract electrode polarization impedance from total measured impedance. The corrected impedance can then be related to the material permittivity. These steps were followed and all measurements were corrected for electrode polarization. The results of the correction procedure are illustrated in Figure B2 for an aqueous solution (EC = 0.67 S/m), and for wet Woodbridge soil saturated with high EC solution (EC = 1.551 S/m). The estimated system parameters for the probe were $R = 0$ and L and C_f vary with frequency. For the simulated frequency range from 1 MHz to 1.5 GHz, L varied from 2.65×10^{-12} (H) to 1.22×10^{-15} (H) and C_f varied from 9.878×10^{-13} (F) to 1.21×10^{-14} (F).

[55] The results clearly show that the effect of electrode polarization for both aqueous solution and soil is limited to low measurement frequency (<20 MHz).

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